

## Project Executable Files

|               |                                                                                    |
|---------------|------------------------------------------------------------------------------------|
| Date          | 15-06-2026                                                                         |
| Team ID       | XXXX                                                                               |
| Project Name  | OptiCrop: Smart Agricultural Production Optimization Engine Using Machine Learning |
| Maximum Marks | 2 Marks                                                                            |

### Project Executable Files

This section contains all the executable and deployment files required for the **Smart Agricultural Production Optimization Engine (OptiCrop)** project. The project has been developed using **Python, Machine Learning, and Flask**. All necessary source code, trained model files, datasets, dependencies, and application files are organized properly so that the evaluator can execute the project successfully.

### Step 1: Submission Checklist

| S.No | Item to Submit                                              | Submitted (Yes / No / NA) |
|------|-------------------------------------------------------------|---------------------------|
| 1    | Complete source code (all files and folders)                | Yes                       |
| 2    | README / Setup Guide (instructions to run the project)      | Yes                       |
| 3    | requirements.txt (dependency file)                          | Yes                       |
| 4    | Dataset (Crop Recommendation Dataset)                       | Yes                       |
| 5    | Environment configuration file (.env.example if applicable) | NA                        |
| 6    | Deployed application URL (if hosted)                        | No                        |

|    |                                                |     |
|----|------------------------------------------------|-----|
| 7  | Executable application (Flask Web Application) | Yes |
| 8  | Dockerfile / Containerization configuration    | NA  |
| 9  | Test files and prediction results              | Yes |
| 10 | Demo video / Walkthrough                       | Yes |

Step 2: File / Folder Structure

The Smart Agricultural Production Optimization Engine (OptiCrop) crop project follows a modular folder structure to separate datasets, machine learning models, frontend resources, backend logic, and supporting files. This organization improves maintainability, readability, and scalability while allowing each component to perform a specific role within the application.

```
OptiCrop/ |
├── dataset/
│   └── crop_recommendation.csv
├── models/
│   └── crop_model.pkl
├── static/
│   ├── css/
│   ├── js/
│   └── images/
├── templates/
│   └── index.html
├── app.py
├── model_training.py
├── preprocessing.py
├── requirements.txt
├── README.md
└── recommendation.py
```

### Step 3: Deployment / Access Details

| Field                       | Details                          |
|-----------------------------|----------------------------------|
| Hosted / Deployed URL       | Not Deployed                     |
| Login Credentials (Demo)    | Not Required                     |
| Platform / Hosting Provider | Local Flask Server               |
| Repository Link             | (Add your GitHub Repository URL) |
| Demo Video Link             | (Add your Demo Video Link)       |

### Step 4: Run Instructions

Follow the steps below to execute the project locally:

1. Download or clone the project repository.
2. Open the project folder in Visual Studio Code or PyCharm.
3. Create and activate a Python virtual environment (optional).
4. Install all required dependencies using:  
`pip install -r requirements.txt`
5. Ensure the trained model file and dataset are available in the project directory.
6. Start the Flask application using:  
`python app.py`
7. Open the browser and visit:  
`http://127.0.0.1:5000`
8. Enter the soil and environmental parameters such as Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, pH level, and rainfall.
9. Click the Predict button to view the recommened crop.

## Step 5: Known Issues / Limitations

| S.No | Known Issue / Limitation                                                                                                              | Workaround / Status                                                                               |
|------|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| 1    | Crop recommendation accuracy depends on the quality and size of the agricultural dataset.                                             | Use a larger and updated agricultural dataset.                                                    |
| 2    | The application supports local deployment only.                                                                                       | Can be deployed on cloud platforms such as Render or Heroku.                                      |
| 3    | The system recommends crops based only on the provided input parameters and does not consider real-time weather or market conditions. | Future versions can integrate live weather APIs and market price data for better recommendations. |